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Portfolio: jagankswain.vercel.app



Profile

Electronics and Communication Engineering undergraduate with a strong focus on **AI/ML systems, VLSI fundamentals, and full-stack development**. Experienced in building end-to-end solutions spanning embedded hardware, data-driven machine learning models, and scalable web applications. Adept at translating complex problem statements into deployable technical systems.

Education

B.Tech – Electronics and Communication Engineering

NIST University, Berhampur

2024 – 2028

CGPA: 9.1

BS – Data Science and Applications

Indian Institute of Technology, Madras

2025 – 2028

CGPA: 8.1

Technical Skills

Programming Languages: C, Python, Java, HTML, CSS, JavaScript, Verilog

Core Computing Concepts: Data Structures & Algorithms, Machine Learning and Deep Learning, Neural Networks, NLP, DevOps

Embedded & Hardware: IoT, Arduino, Raspberry Pi, Digital Signal Processing, FPGA, EDA Tools, VLSI (Verilog)

Tools & Platforms: Linux, PyCharm, MATLAB, Google Colab, Vivado, AWS, Google Cloud Platform

Mentor

Dr. Sandipan Mallik

Professor, Department of Electronics and Communication Engineering

NIST University, Berhampur



Projects

GuardX Sentinel – Drone-based Human Activity Detection System

Computer Vision, IoT, Embedded Systems

- Developed an indigenous drone-integrated surveillance system for real-time human activity detection and analysis.
- Implemented computer vision and machine learning algorithms for object detection and threat identification in defense scenarios.
- Integrated Arduino Nicla Vision, camera modules(Thermal and Regular), and onboard sensors to enable efficient edge processing and real-time data acquisition.
- Optimized system performance for low-latency operation under constrained computational resources.
- Awarded **Best Innovation Award** at Chakravayuh 1.0 and received a **Rs.50,000 Defence R&D Grant** at the International Conference on Air Defence and Security.

SegroBin – Smart Waste Management System (SIH 2025)

IoT, Embedded Systems, Application Development

- Designed a smart waste segregation system addressing Government of Odisha problem statement SIH25046 and were the Top 5 Finalists from the given PS.
- Developed an IoT-enabled smart dustbin integrated with a companion application for real-time waste tracking.
- Implemented incentive mechanisms using Green Credits to encourage recyclable waste segregation.
- Engineered a low-cost, scalable solution suitable for municipal waste collection and recycling workflows.

16-Operation ALU Design using Verilog HDL

Verilog, Digital Design, VLSI

- Designed and implemented a 16-operation Arithmetic Logic Unit (ALU) using Verilog HDL as the core computational module of a processor.
- Supported arithmetic operations (+, -, *, /), logical operations (AND, OR, XOR), shift operations, and comparison functionalities.
- Implemented status flags including Carry (C) and Zero (Z) for conditional execution and control flow.
- Performed functional and timing simulations using Vivado 2025.2, achieving accurate waveform verification.

Formula 1 Race Winner Prediction System

Machine Learning, Time-Series Analysis

- Built predictive models using historical Formula 1 race data sourced from Kaggle.
- Performed extensive feature engineering, capturing driver form, season momentum, circuit affinity, and grid position.
- Trained and evaluated Logistic Regression, Random Forest, Gradient Boosting, and Neural Network models.
- Achieved up to 95.6% cross-validated accuracy using Neural Networks with time-aware validation.

DeepBeat – ECG Arrhythmia Classification System

Deep Learning, Medical Time-Series Analysis

- Developed a hybrid CNN-LSTM model in PyTorch for cardiac arrhythmia detection.
- Trained on the MIT-BIH Arrhythmia dataset with over 87,000 ECG samples.
- Achieved 96% overall accuracy with high recall for critical ventricular and supraventricular classes.

- Designed the system as a safety-first clinical screening support tool.

AI Career Agent – GenAI Hackathon Project

AI Agents, Full-Stack Development

- Built an AI-driven career guidance agent for the IndiaAI Impact GenAI Hackathon (IISC Bengaluru in collaboration with MeitY, Govt. of India).
- Implemented skill extraction, job-role matching, and personalized learning path generation.
- Developed a modular architecture with a Flask backend, AI reasoning engine, and SPA frontend.
- Leveraged IBM open-source technologies to implement agentic workflows beyond keyword matching.

Full-Stack Authentication System

Node.js, Express, MongoDB

- Designed and implemented a secure authentication system from scratch.
- Integrated JWT-based authentication, bcrypt password hashing, and input validation.
- Built a responsive frontend with real-time form validation and user feedback.

Smart Blind Stick

Embedded Systems, IoT

- Developed an assistive navigation device using sensors and Arduino for obstacle detection.
- Implemented real-time alert mechanisms to enhance user safety.

Arduino-Based 3D Scanner

Embedded Systems, Computer Vision

- Designed a rotating scanning platform using stepper motors and microcontroller control.
- Coordinated hardware motion with camera-based image acquisition for 3D reconstruction.

Achievements

- Recipient – **Rs.50,000 Defence R&D Grant**, awarded by Army AD College & NIST University at the International Conference on Air Defence and Security for the project **GuardX Sentinel**.
- Winner – **Best Innovation Award**, Chakravyuh 1.0, for the project **GuardX Sentinel**, a drone-based human activity detection system.
- Grand Finalist – Smart India Hackathon (SIH) 2025 (Hardware Edition), as part of Team **CtrlOhm**, B.Tech 2nd Year cohort.
- 1st Place – **Web Development Competition**, Festronix 2K26, GIFT Autonomous, Bhubaneswar; demonstrated outstanding technical and development skills.
- Winner – **Code Crusade 4.0** (Group 2: 1st & 2nd Years), organized by Club Excel, NIST University; competitive coding contest conducted on HackerRank.

Certifications & Training

- **OOSC 3.0 – Open-Source Software Conference**, IIT Kanpur
Conducted by Canonical and Ubuntu; exposure to Linux, open-source ecosystems, and collaborative development practices.
- **DevOps Webinar** – Malla Reddy Engineering College & Google Developer Groups (GDG)
Learned DevOps principles, CI/CD pipelines, and modern development workflows bridging development and operations.

- **IoT and Mechatronics Internship** – NIST University
Hands-on experience with Arduino-based embedded systems, sensors, automation, and cloud-integrated IoT solutions.
- **Build and Deploy Machine Learning Solutions on Vertex AI** – Google Cloud Skill Badge
Practical experience in building, training, and deploying ML models using Google Vertex AI.
- **Data Analytics Job Simulation** – Deloitte Australia (Forage)
Completed a professional simulation involving data analysis and forensic technology; built dashboards using Tableau and derived business insights using Excel.
- **Blender and AR Workshop – The Dream Experience**
Conducted by Club Multimedia, NIST University; hands-on experience with Blender fundamentals, Unity, and AR application development.
- **Internet of Things Internship** – LaunchED Global
Worked with sensors and actuators, Arduino Cloud, Blynk, TinkerCAD, OTA updates, and Wi-Fi-based IoT communication.
- **SensoTech Workshop** – Renewable Energy Club, NIST University
Two-day workshop on IoT sensors and actuators; implemented mini-projects including RFID-based attendance system and rain detection alert system.
- **International Tech Event Certification** – Google & Google Developer Groups
Gained insights into Vertex AI and Gemini applications, including real-world visualization projects showcased at Sphere, Las Vegas.
- **AI Cloudwave** – Cloud Computing Club, NIST University (Nov 2024)
Exposure to AI and cloud technologies including AWS, Generative AI tools, Google Gemini, Claude, Sonnet, and Haiku.